

RELRAD - AN ANALYTICAL APPROACH FOR DISTRIBUTION SYSTEM RELIABILITY ASSESSMENT

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Trondheim, Norway**Abstract** - The paper addresses:

- The use of reliability analysis within the framework of distribution system planning.
- A specific model for reliability calculations in radially operated distribution systems.

Due to increased outage costs in terms of specific costs of non-delivered power and energy (NDP and NDE), an increased part of system operating costs is related to reliability aspects.

Customer's outage costs, based on the willingness-to-pay principle, are given the same principal treatment as power system electrical losses. This approach is becoming generally accepted by the power companies in Norway.

The reliability model is based on an analytical approach, connecting component failures to load point outages. The description is based on statistical distributions.

Two case studies are reported in the paper.

Key words: System planning, distribution systems, reliability analysis, computer model.

1. INTRODUCTION

Traditionally, reliability analysis of electrical distribution systems was considered as a tool for the planning engineer to ensure a 'reasonable' quality of service and to choose between different system expansion plans that costwise were comparable considering system investments and cost of losses. In some cases the cost of non-delivered energy and power (NDE, NDP) were included in the planning objective, but this approach was hampered by disputes concerning the levels of specific values of NDE in various sectors.

Recently, a new Energy Law has been approved by the Norwegian Government, urging for a more market based purchase of electrical energy. The Law allows for a deregulated and a competitive structure in the energy markets. In the framework of distribution system planning these new terms of delivery combined with recent developments in metering technology are leading to:

- An increased interest in demand-side management by variation in quality of service.
- An increased acceptance of cost estimates of customers' willingness-to-pay for quality of service.

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From this, it is obvious that the utilities have to pay more attention to reliability analysis of the distribution systems in order to ensure a quality of service that is in accordance with the customers' quality expectations reflected by their willingness-to-pay.

This paper provides an overview of the distribution system planning model used by Norwegian utilities with emphasis on the approach used for reliability analysis, RELRAD (RELIability in RADial systems). The use of RELRAD is placed within the framework of distribution system planning, and two case studies of typical Norwegian MV systems are included:

- Benefit of automation and remote control in a rural overhead system.
- Impact of reserve cable in an urban cable system.

2. RELIABILITY AS PART OF DISTRIBUTION SYSTEM PLANNING

The planning procedure:

The planning process is dealt with in six main steps:

1. Data basis evaluation
2. Outline analysis/System philosophy considerations
3. Technical simulations
4. Cost evaluation
5. Cost optimization
6. Result evaluation

Steps 1-5 are phases with a strong interplay between the planning engineer and a distribution system planning and documentation program, called NETBAS [10], while step 6 is largely a manual phase involving engineering judgement and experience.

Data basis evaluation comprises modelling of the existing distribution system and loads by data retrieval from the documentation database, specifications for future loads and possible future system investments, specification of various economic parameters and restrictions.

Outline analysis involves general studies where a large number of system solutions are considered. A stylized network description is used with simplified and regular structure regarding both network topology and loads. The computer programs automatically generate a large number of system solutions (~10 000 in practical cases). Different technical characteristics, such as network losses, NDE, voltage drops etc., are calculated over the period of analysis. These studies are occasionally carried out in order to give guideline signals for the detailed distribution system planning.

Technical simulations are a computer application phase based on interaction between system planner and various computer programs. The main aim is to find which future system solutions are technically qualified to be evaluated in subsequent phases and for which part of the period of analysis they satisfy system restrictions.

Load flow, short-circuit and reliability analyses are carried out to evaluate the technical performance of the conceivable future solutions.

Sections 3 and 5 give a detailed description of the reliability part of the technical simulations.

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Cost evaluation/Cost optimization. When fixed costs (investments) and operating costs (costs of system losses and NDE/NDP) are evaluated for all qualified system solutions throughout the period of analysis, the remaining task is to determine the plan of development that contributes to the optimization of expected total costs.

The main optimization techniques utilized in this phase are Dynamic Programming and Mixed-Integer Programming.

3. THE RELIABILITY CONCEPT

Figure 1 shows the reliability concept used by EFI.

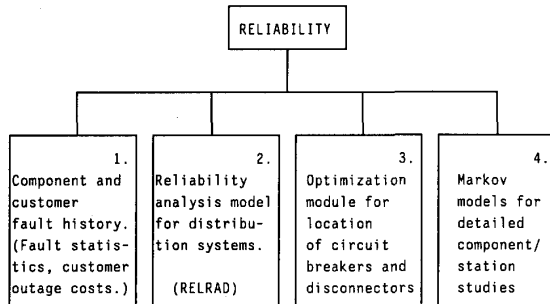


Figure 1. Reliability concept.

Component and Customer Fault History

This module is intended for registration, administration, storage and report generation of both component fault history and statistics, and customer fault history, statistics and interruption cost. The function of the module is twofold:

- providing data for other modules (reliability analysis, maintenance module etc.).
- recording customers' quality of service.

Reliability Analysis Module

This module is the main analysis tool for distribution system reliability analysis and is further outlined in Section 5.

Optimization of Breaker Equipment

This module is at an experimental stage. The location of circuit breakers and disconnectors is heavily influencing the quality of service especially in overhead line distribution systems (rural areas).

The applied method is an extension of the method proposed in [11].

Markov Models

This module is used for detailed analysis of different failure modes of components in order to:

- Analyse the effect of various counter measures against failure mechanisms.
- Provide equivalent fault statistics for complex equipment, i.e. stations etc.
- Provide 'fault statistics' for new components with a limited number of years in service.

4. RELIABILITY EVALUATION TECHNIQUES

Models to evaluate reliability indices for distribution systems have been the subject of many papers (e.g. [1], [3],[4]). There are three basic indices common in use: The expected failure rate (λ), the average outage time (r) and the expected annual outage time (U). These are adequate for simple radial systems and more extended indices are developed for general distribution systems (mixed radial and meshed systems), based on both a total and a partial loss of continuity criterion (e.g. [2]).

To evaluate cost-benefit of reinforcements and measures to improve reliability, two additional indices are being used: The expected annual NDP and NDE (e.g. [2], [3], [5], [6]).

There seem to be two main approaches applied to reliability evaluation of distribution systems which have been discussed previously in many papers:

- I. Simulation methods based on drawings from statistical distributions (Monte Carlo).
- II. Analytical methods based on solutions of mathematical models.

The Monte Carlo techniques are normally 'very' time-consuming due to the large number of drawings necessary in order to obtain accurate results. The fault contribution from each component is given by a statistical distribution of failure rates and outage times. One of the advantages of simulation methods is that no assumptions have to be made concerning the shape of the statistical distributions applied.

The analytical approach is based on assumptions concerning the statistical distributions of failure rates and repair times. The most common evaluation techniques, using a set of approximate equations, are failure mode analysis or minimum cut set analysis. The methods applied are less time-consuming than the simulation methods, but suffers from the problem of representing repair times adequately.

Further results obtained by simulation are distributions associated with the reliability indices. Analytical approaches to evaluate such distributions are proposed (e.g. [8],[9]).

The next section of this paper describes an analytical approach to reliability evaluation of radial distribution systems. The approach is called RELRAD and is the complementary to the minimum cut set approach.

5. THE RELRAD MODEL

5.1 Method Description

The computer based method RELRAD calculates individual load point reliability indices in radially operated MV distribution systems.

The method has its origin in a previous simulation model, developed at EFI, and started as an improvement of the topology representation in the simulation model. In order to reduce the computation time, a method for determination of expected values was developed (RELRAD). This method has been further improved and extended as will be described in the following.

RELRAD is an analytical approach, based on the fault contribution from all network components and their consequences to the load points outages. This differs from failure mode or minimum cut set analysis which assesses the individual load points reliability directly by the minimum cut set. In short the minimum cut set

analyses the individual load points, while the analytical simulation approach analyses the individual network components. This is shown in Fig. 2.

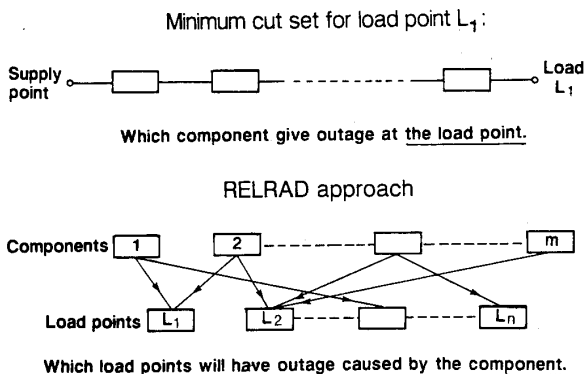


Figure 2. Analytical techniques for radial systems.

The full network topology is reduced to reliability sections (fig. 3), defined by the location of breaker equipment.

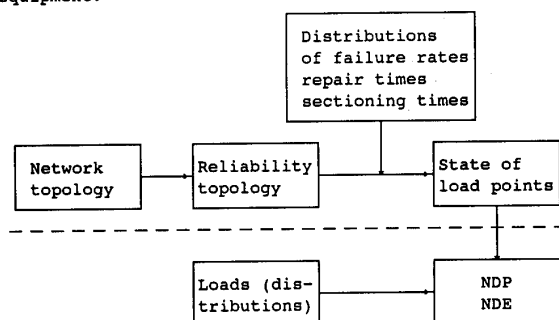


Figure 3. Model for reliability analysis (RELRAD).

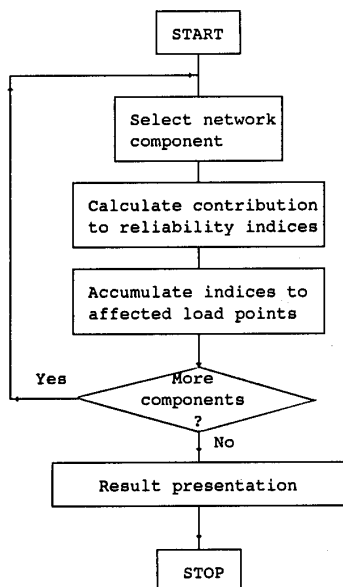


Figure 4. Flow chart for calculation of reliability indices.

The depth of system load points' outages are determined by the circuit breakers' locations. Some disconnected loads will have the supply restored after a short sectioning time while others will be connected after the repair of the faulty component.

The logic describing the relation between the components and their fault contribution to the different load points - the reliability topology - is generated before starting the reliability calculations.

The next step is to generate expected values from the statistical distributions of failure rates, repair times and sectioning times for all components in the system.

The algorithm then accumulates reliability indices for each load point from each component giving outages to the load point. Finally, when the fault contribution from every component is investigated, the total accumulated indices are available.

Fig. 4 shows a flow chart of the algorithm.

5.2 Main Assumptions

The topology description is based on the following assumptions:

- radial operation of the network
- no transfer restrictions on reserve connections
- all faults are isolated by the upstream circuit breakers, by the first or the second depending on the probability of malfunction of the circuit breaker
- when the fault is located, the upstream disconnector will be opened and the circuit breaker closed.

Components represented in the model:

Lines
Cables
Main transformers
Circuit breakers
Control equipment
Disconnectors
Distribution transformers

Main statistical assumptions:

- All failures are statistically independent.
- Multiple faults are not represented except for circuit breaker malfunction.
- All failures are repaired before next fault occurs.

5.3 Other Features of the Model

The main interest so far in reliability analyses has been to consider forced outages leading to repair due to the use of NDE as a measure of customers' inconveniences of outages. In addition to permanent faults, temporary faults and voltage dips are very important to certain customers giving significant costs of NDP. Scheduled outages will also lead to relatively large inconvenience costs compared to forced outages. The RELRAD model can handle all three types of outages by using specific failure statistics for forced outages, temporary faults or scheduled outages.

The reliability calculations may in addition to the mentioned components, take into account different reserve possibilities, e.g.

- normally open ring connections within the network
- mobile reserve cables

- mobile generating units
- mobile substations or other external reserve supply

Automatic sectioning devices or remote control on certain disconnectors may be specified by reduced sectioning times for these disconnectors/circuit breakers.

5.4 Reliability Indices Calculated by the Model

The following reliability indices (expected values) are evaluated:

$$f = \sum_i \lambda_i \quad (= \lambda) \quad \text{[outages/year]} \quad (1)$$

$$\Delta T = \sum_i \lambda_i k_i \quad (= U) \quad \text{[hours/year]} \quad (2)$$

$$r = \Delta T / f \quad \text{[hours/outage]} \quad (3)$$

where

- f - annual frequency of outages
- ΔT - annual duration of outages
- r - average outage time
- λ_i - expected no. of faults per year for component no. i
- k_i - average repair or sectioning time for component no. i

These indices are not deterministic values, but expected or average values of underlying probability distributions and represent long-run average values.

In addition the program calculates expected NDP and expected NDE for each load point:

$$\Delta P = P \cdot f \quad \text{[kW-outages/year]} \quad (4)$$

$$\Delta W = P \cdot \Delta T \quad \text{[kWh/year]} \quad (5)$$

where P - power [kW]

The algorithm calculates reliability indices, in terms of mean values and distributions for the number of outages and annual duration of outages. Indices are calculated both for the individual load points and for the whole system.

When specific costs of NDE and NDP are given, the program calculates total outage costs.

For certain purposes the probability distributions of number and duration of outages are of interest, in order to evaluate for instance 50 % and 90 % probability values.

In RELRAD the number of outages are assumed to follow the Poisson distribution, and the relevant distribution parameters are calculated from the failure statistics.

Due to the problems by representing repair times by any known distribution for analytical calculation, the following approach based on the statistical distributions is used in RELRAD:

$$F_i(t) = \frac{\sum_j F_{rj}(t) \cdot \lambda_i + \sum_j F_{kj}(t) \cdot \lambda_j}{\sum_i \lambda_i + \sum_j \lambda_j} \quad (6)$$

where

- $F_i(t)$ - cumulative distribution for outage duration at load point No. 1
- $F_{rj}(t)$ - statistical cumulative distribution of repair time for component No. i
- $F_{kj}(t)$ - statistical cumulative distribution of sectioning time for component No. j
- λ_i - expected no. of faults per year for component No. i leading to outage duration equal to repair time for load point No. 1

- λ_j - expected no. of faults per year for component No. j leading to outage duration equal to sectioning time for load point No. 1.

The components are divided in two parts: those leading to outage time equal to repair time, and those leading to sectioning time at the load point.

The RELRAD method gives the same expected values as the original simulation model while the computation time is reduced by a factor of 10-100 depending on the network size and the simulation period.

6. CASE STUDIES

6.1 Failure Statistics

To perform reliability analyses, either the utilities' local statistics or average values for regions or the country as a whole, can be used as input.

Average expected failure rates and repair times for the two cases are shown in Table 1. Sectioning times will vary with types of disconnectors and control equipment and within the utilities supply area, due to distances and geographical/climatical differences. Sectioning times may therefore be given as specific outage times for each disconnector or as an average for the system.

TABLE 1. Failure statistics (for permanent faults), average values.

Type of Component	Failure rate* [No. per year]		Repair time [hours/outage]	
	Case 1	Case 2	Case 1	Case 2
Lines	6.0	-	2.0	2.0
Cables	5.0	4.7	13.7	13.7
Circuit breakers	1.7	1.3	6.0	5.4
Control equipment	5.0	1.7	0.8	0.8
Disconnector	1.4	0.07	1.5	0.8
Distribution transf.	1.0	0.18	3.0	7.8

*Per 100 units. The unit is km for lines and cables.

Failure rate, repair time and load will vary through the year. Annual distributions of these parameters will show the correlation between them. In Norway, the failure rate tends to be high in summer and winter, while the repair time is high during winter. The peak load periods will also occur in winter. The three parameters mentioned show a strong positive correlation in the winter period, hence influencing greatly on the annual values of NDE and NDP. Therefore it is not correct to use average or peak power values for the calculation of NDP and NDE (referring to eq. 4 and 5). Correlation factors are calculated from annual distributions of failure rate, repair time and load.

6.2 Benefit of Automation and Remote Control (Case 1)

The RELRAD program has been used to investigate the impact of remote controlled disconnectors and automatic sectioning devices on the overall reliability of a typical rural area overhead MV system.

The network consists of 66 km overhead lines supplying 82 distribution transformers. Average expected sectioning time is 50 minutes per interruption for this area. Remote control of the disconnectors will reduce the interruption time for the sound parts of the network to 10 minutes. This investment is compared to installation of a circuit breaker at point A in Figure 5. Average sectioning time with manual operation will be 35 minutes while the circuit breaker in combination with remote control or automation will reduce the sectioning time to

6 and 1 minute respectively. There are no reserve connections in the network.

Table 2 gives a description of the 5 analysed alternatives.

Table 2. Alternatives for case 1.

Alternative	Measure		
	Circuit breaker	Remote control	Automatic sectioning
1. Existing network			
2. Remote control with fault indicators(FI)		X	
3. Circuit breaker at A	X		
4. Circuit breaker at A and remote control with FI	X	X	
5. Equals case 4 + computer equipment for automatic sectioning	X	X	X

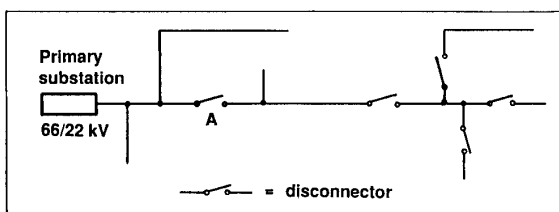


Figure 5. A 22 kV MV radial.

Table 3 shows expected values and annual cost of investments for the different alternatives. Failure rates and repair times are given in Table 1.

Table 3 gives figures for forced outages. In this network forced outages represent 29 % of the total number of faults and 48 % of the NDE. The corresponding values for temporary faults are 41 % and 12 % respectively, and for scheduled (planned) outages 30 % and 40 % respectively.

Temporary faults are defined as faults not leading to repair. Such faults are normally removed by the automatic reclosing procedure of the circuit breaker. As the model comprises failures of the reclosing equipment, temporary faults will also contribute to the NDE.

Table 4 gives figures for the three types of outages for the radial in Figure 5.

The expected number of outages per year is reduced from 5 to 3.8 by installing the circuit breaker, while the annual outage time is reduced by two hours with automatic sectioning of all breakers. In this alternative, which is the best from a reliability point of view, the NDE is reduced by approximately 30 %.

The break-even cost in Table 3 is the specific cost of NDE that makes the investment cost-effective. Alternative 3 has the lowest break-even cost and represents a relatively small investment giving reductions in both number of outages and outage duration.

Automatic sectioning (Alt. 5) is cost-effective for any cost of NDE higher than 6.4 NOK¹⁾/kWh. This cost of NDE

1) 1 US\$ ≈ 6 NOK.

TABLE 3. Average expected outage values and investment costs for case 1.

Alt.	No. of outages per year	Annual outage time [hours]	NDE 1988 [kWh/year]	Annual cost of invest.*) [NOK/year]	Break-even cost NDE [NOK/kWh]
1	5.0	6.5	28 900		
2	5.0	4.8	21 000	39 000	4.9
3	3.8	5.3	23 100	22 000	3.8
4	3.8	4.6	19 900	55 200	6.1
5	3.8	4.5	19 400	60 700	6.4

*) Assumed economic lifetime is 15 years and rate of interest is 7 %.

Table 4. Average expected outage values, different types of outages, case 1.

Outage	No. of outages per year	Annual outage time [hours]	NDP kW, 1988	NDE kWh, 1988
Forced	5.0	6.5	22 300	28 900
Temporary	7.0	1.5	31 500	7 225
Scheduled	5.2	9.0	23 000	24 100
Average for the network	17.2	17.0	76 800	60 225

is quite small compared to the assumed average customers' interruption cost of 9 NOK/kWh for this area.

6.3 Impact of Reserve Cable (Case 2)

The 11 kV system in Figure 6 is a typical urban area cable system in Norway. There are nine outgoing cables (7 km in total) from the busbar, supplying 25 distribution transformers. Cable No. 8 carries no load and serves only as a reserve for the other cables. Except

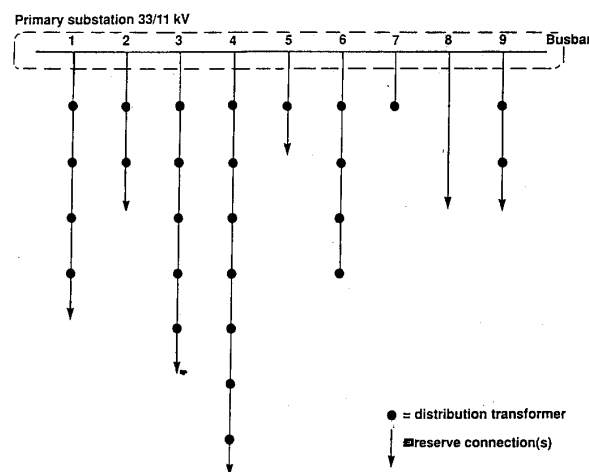


Figure 6. An 11 kV cable system.

for cables Nos. 6 and 7, all the cables have other reserve possibilities in addition to cable No. 8. This is indicated by an arrow on the figure.

To investigate the impact of the unloaded reserve, reliability indices with and without cable No. 8 are calculated. Failure rates and outage times are given in Table 1. Average expected sectioning time is 2 hours 20 min. The results are shown in Table 5:

Table 5. Average expected outage values, NDP and NDE for the system in Figure 6. (Case 2.)

Unloaded reserve	No. of outages per year	Annual outage time [hours]	NDP [kW/year]	NDE [kWh/year]
With	0.20	0.54	2220	5980
Without	0.19	0.57	2070	6250

Without the unloaded reserve cable, average number of outages and NDP would be reduced, while annual outage

time and NDE would increase. The reduction and increase are both small (approx. 5 %) due to the fact that most of the cables have additional reserve possibilities.

With cost of NDP equal to 2 NOK/kW and 20 NOK/kWh for NDE, the unloaded reserve cable represents a reduction in annual outage costs of NOK 5100. Annual cost of equipment and installation of this cable amounts to NOK 40 000. Hence, the unloaded reserve cable is not cost-effective.

7. CONCLUSIONS

Due to changing premises and requirements, reliability analysis is playing an increasingly important part of power system planning, especially in MV distribution systems.

The model developed so far is deemed suitable for the present level of use and available failure statistics. The model compared with the simulation type of models, gives the same expected values.

Further developments have to be based on experience from the utilities using the model and improvement of fault statistics.

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Her main activities have been in the distribution system planning area, reliability in particular.